ENN554 Project Manual

Design of a Wind Microgrid

Faculty of Engineering

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1 Project Brief

In this project you will design a wind farm and size a battery to satisfy an electrical load. Your wind farm will be at a pre-selected location offshore in the Gippsland Offshore Renewable Energy Infrastructure Region. As shown in Figure 1, the wind farm must be located in bounding box shown, which represents a 300 hectare ($3\,\mathrm{km}^2$) square centred at a latitude of -38.4° and longitude of 147.4°. In practise, designers must consider the statistics of the

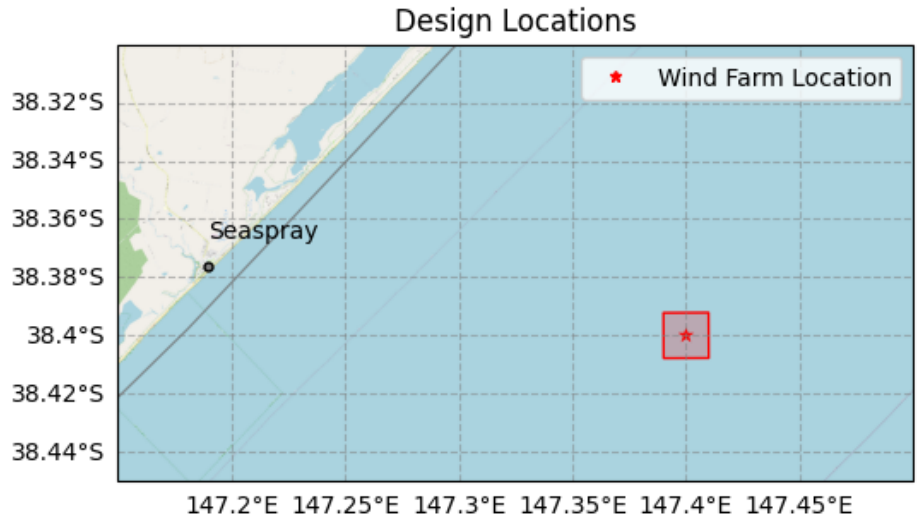

Figure 1: Detailed view of the wind farm location.

load and renewable resources to ensure that the load can be satisfied at all times (e.g. under peak demand or and/or cloudy days). However, **such analysis is outside the scope of this assessment, and we will instead design the system on a single year alone.** The load data is posted on Canvas with this manual and is shown in Fig. 2. The data are for one year, which is assumed to be "typical" so that we can use it for design.

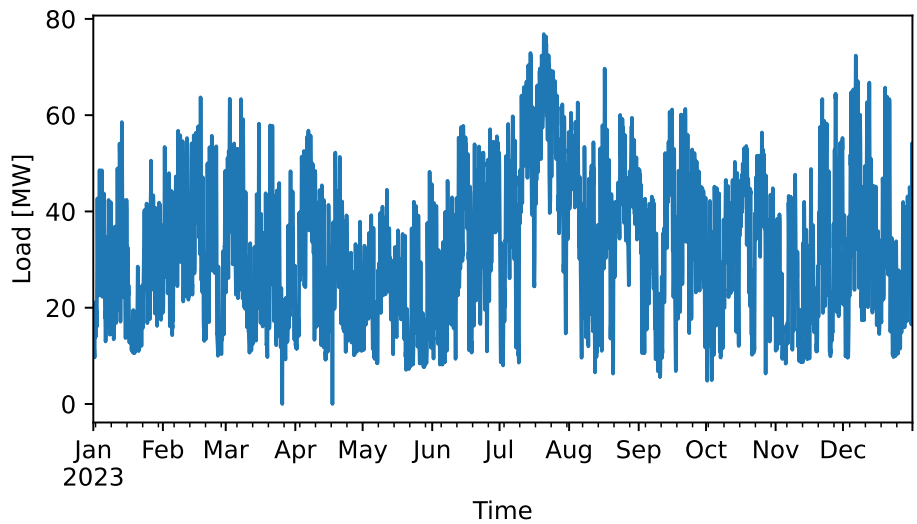

Figure 2: Load to be satisfied by the PV-Wind system. The data is available in the `load_project_2025.xlsx` fine on Canvas.

2 Wind Resources

Your first task is to characterise the wind resources at the sites. The wind data may be obtained from the NASA POWER website using the Single Point option on the left and the centre of the wind farm bounding box, which was provided in the previous section. You should characterise the wind speeds and directions using typical tools from class (e.g. a wind rose). Be sure to include the plots in your write up and discuss their implications for your later design.

3 Select the wind turbine

In this section, you will select a turbine using the annual energy production (AEP) of the single turbine. The AEP can be computed using a Weibull wind speed model[1] and assuming that the turbine is always facing into th wind. The formula for AEP with the Weibull model is

$$AEP = 8760 \cdot \int_0^\infty P(U)\, f(U)\, dU \tag{1}$$

where $f(U)$ and $F(U)$ are, respectively, the probability density function (PDF) and cumulative distribution function (CDF) of a Weibull distribution, and $P(U)$ is the power curve

[1]Note that the use of this model implies that we are ignoring hour-to-hour autocorrelation. This simplifying assumption seems to be quite common, even if autocorrelation is likely present.

of the wind turbine. The total capital cost of the wind turbine can be calculated using the following equation

$$TCC = (C_{rotor} + C_{tower})\, A_{rotor} + C_{other} P_{rated} \tag{2}$$

where C_{rotor} and C_{tower} are the rotor and tower costs (units: $\$/m^2$), respectively, while C_{other} accounts for other costs (e.g nacelle) that scale with the rated power (units:$\$/kW$). Refer to Table 3 for values.

You were provided the *power curves and cost data* for four (4) turbines with this assignment description from NLR's database. Compute the AEP using (1) and the LCOE for a single-turbine wind farm. Be sure to motivate your selection of the 5 candidate turbines and justify any choices (e.g. the discount rate). Include any appropriate plots and tables to justify your choices and include the following table:

Table 1: Table format for displaying the LCOE results.

	Annual Energy Production (GWh)	Rotor + Tower cost (AUD)	Other costs (AUD)	TCC (AUD)	LCOE (AUD/MWh)
Turbine 1					
Turbine 2					
Turbine 3					
Turbine 4					

4 Wind farm production

Select a turbine and develop the layouts for a 9, 16, and 25 turbine farm. Be sure to describe the logic of your layout method since your mark will be based on the quality of your reasoning and solution.

5 Microgrid design on energy basis

In this section, you will design a microgrid so that the electricity demand is always supplied by either the wind turbines or a battery energy storage system (BESS).

Several microgrid configurations will be considered. Specifically, each wind farm layout from the previous section is coupled with a BESS with maximum storage capacity $E_{st,max}$.

The design procedure relies on a simple energy balance that must hold for every hour i of the year. The two unknowns of the problem are i) the **minimum** number of PV sub-arrays required to meet the load, and ii) the corresponding **minimum** battery storage size.

$$\Delta E_{st,i} = E_{WF,i} - E_{load,i} \tag{3}$$

Here $E_{pr,i}$ is the total electricity production of the microgrid, $E_{WF,i}$ is the hourly production of the wind farm and $\Delta E_{st,i}$ is the net surplus (positive) or deficit (negative) of electricity relative to the load $E_{load,i}$. Your task here is to size the battery so that the load is satisfied for every hour available. Discuss your methodology and results in your report and fill in the following table

Table 2: Table format for displaying the micro-grid design results.

	Wind farm nameplate capacity (MW)	BESS capacity (MWh)
Wind farm 1		
Wind farm 2		
Wind farm 3		

The optimal microgrid configuration is identified among based on their economic performance. The key performance indicator adopted is the **Actual Cost of Electricity (ACOE)**, which is conceptually similar to the LCOE but also accounts for grid constraints on electricity production and curtailed energy. In the present case, the actual electricity production coincides with the load demand.

The ACOE of the microgrid is defined as:

$$ACOE = \frac{FCR \cdot TCC + OC}{\sum_{i=1}^{8760} E_{load,i}} \tag{4}$$

where TCC is the total capital cost of the system, obtained by summing the contributions of each component:

$$TCC = C_{WF} + C_{BESS} \tag{5}$$

OC represents the operating costs of the microgrid, including both fixed and variable components. For simplicity, variable operating costs (VOC) are assumed negligible, so that OC accounts for fixed operating costs (FOC) only:

$$OC = FOC_{WF} + FOC_{BESS} \tag{6}$$

The wind farm capital cost, C_{WF} can be computed by multiplying the single turbine cost in Section **??** by the number of turbines included in the considered layout. BESS capital costs C_{BESS} can be computed as:

$$C_{BESS} = \max\left(c^{en}_{BESS} \cdot E_{st,max}; c^{pow}_{BESS} \cdot P_{st,max}\right) \tag{7}$$

Here, $E_{st,max}$ and $P_{st,max}$ denote the maximum energy and power capacity of the BESS, respectively. Finally, the fixed charge rate, FCR, scales the TCC accounting for plant lifetime, construction time, nominal return of investment and inflation rate. Refer to Table 3 for cost components descriptions and assumptions.

Select the best wind farm and BESS combination. Discuss your choices and detail your analysis in your report.

A Cost data for economic analyses

Table 3: Cost assumptions

Symbol	Description	Value	Units
c_{rotor}	WT rotor cost	170	A\$/(m^2 swept area)
c_{tower}	WT tower cost	90	A\$/(m^2 swept area)
c_{other}	WT other cost (e.g. nacelle)	1000	A\$/kW
c^{en}_{BESS}	BESS specific cost on energy base	300	A\$/kWh
c^{pow}_{BESS}	BESS specific cost on power base	375	A\$/kW
FOC_{WF}	Wind farm fixed OC cost	60	A\$/kW
FOC_{BESS}	BESS fixed OC cost	10	A\$/kWh
FCR	Fixed charge rate	You choose!	[-]